

Technology Stack

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	2 Marks

Define Your Technology Stack

Every entry below is the technology actually committed to the repository and pinned in requirements.txt, along with the reasoning already recorded in the project's own engineering notes — not a generic stack description.

S.No	Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit 1.37.1	Rapid, Python-native UI well suited to a data/ML-heavy app; avoids building and maintaining a separate JS frontend.
2	Emotion Model 1	TensorFlow (tensorflow-cpu 2.16.1) — BiLSTM	Small, fast baseline model. CPU-only is intentional, not a compromise: TensorFlow dropped native Windows GPU support after v2.10, and CPU training was actually faster on a Ryzen 9 9900X than relying on Kaggle notebooks.
3	Emotion Model 2	PyTorch + Transformers 4.44.2 — fine-tuned BERT	Stronger contextual understanding than BiLSTM; fine-tuned specifically on the 5-class emotion set, GPU-accelerated locally on an RTX 5070 Ti.
4	LLM Response Generation	google-generativeai 0.8.3 (Gemini) primary; openai 1.51.0 SDK pointed at OpenRouter as an alternate provider, selected via the LLM_PROVIDER env var	Keeps response generation swappable between providers without touching calling code; OpenRouter was added to work around a Gemini-specific issue hit during local development.
5	Model Hosting	Hugging Face Hub (huggingface_hub==0.24.6, pinned below 1.0)	BERT weights are too large for git; downloaded at runtime via HF_MODEL_REPO_ID when local weights aren't present. The version pin exists because upgrading huggingface_hub breaks transformers' import entirely — hit once already.
6	Data Handling & Analytics	pandas 2.2.2, numpy 1.26.4, scikit-learn 1.5.1, Plotly 5.24.1, matplotlib	pandas/numpy for preprocessing and the decision-signal analysis scripts; scikit-learn for classification_report/confusion_matrix; Plotly for the interactive analytics tabs; matplotlib for the static training-curve and confusion-matrix images.
7	NLP Preprocessing	nlTK 3.9.1, contractions 0.1.73	Text cleaning and tokenization ahead of both models.
8	Persistence	CSV file (src/persistence/csv_logger.py) — no database	Lightweight, dependency-free logging suited to a single-session demo/assessment context; documented as not surviving restarts on non-persistent hosting disks.
9	Deployment	Streamlit Community Cloud (public, deliberately degraded) + local full-feature development (Windows, RTX 5070 Ti)	Free-tier public target. BERT and AI response generation are switched off on the public deployment (DISABLE_BERT / DISABLE_AI_RESPONSE) due to an unresolved OpenRouter call hang under that environment; local dev runs the full pipeline.
10	Version Control	Git + GitHub	Standard incremental-commit workflow — 46 commits across the active development window, organized by epic and task.